

James Wedgwood

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SUMMARY

Machine-learning researcher and engineer with 4+ years of industry experience building deployed ML systems and current research in scientific machine learning, probabilistic modeling, and LLM safety. Experienced in developing open-ended research ideas into rigorous methods and usable software.

EDUCATIONAL EXPERIENCE

Carnegie Mellon University — *M.S. in Machine Learning*

EXPECTED DECEMBER 2026

4.08 GPA | Research advised by Virginia Smith and Barnabás Póczos

Selected coursework: Deep Reinforcement Learning and Control, Optimization for Machine Learning, Probabilistic Graphical Models, Advanced Introduction to Machine Learning, Intermediate Statistics

Yale University — *B.S. in Mathematics, Literature & Comparative Cultures*

2016 – 2020

3.95 GPA; *summa cum laude*; Phi Beta Kappa; distinction in both majors

RESEARCH EXPERIENCE AND PUBLICATIONS

Conserved Quantity Detection in Physical Simulations — *Advised by Barnabás Póczos*

2026 – PRESENT

Independently developing an **end-to-end probabilistic framework** using Gaussian processes, Bayesian quadrature, and active sampling to identify conserved quantities from large magnetohydrodynamic simulations

Safety Nudges — *Advised by Virginia Smith*

2026 – PRESENT

Designed and implemented a system that detects **safety risks in human-LLM conversations** and offers context-sensitive interventions, with IRB-approved user-study evaluation led by a collaborator

James Wedgwood, Aashiq Muhamed, Mona T. Diab & Virginia Smith. “DSPA: Dynamic SAE Steering for Data-Efficient Preference Alignment.” *EMNLP Main Conference*, 2026 [\[code\]](#) [\[paper\]](#)

James Wedgwood, Pratiksha Thaker, Neil Kale & Virginia Smith. “Personalization, Personas, and Forecasting in Value Alignment.” *ICML Pluralistic Alignment Workshop*, 2026 [\[paper\]](#)

James Wedgwood, Chhavi Yadav & Virginia Smith. “Automated Concept Discovery for LLM-as-a-Judge Preference Analysis.” *ICLR UCRL Workshop*, 2026 (**Oral**) [\[code\]](#) [\[paper\]](#)

James Wedgwood. “P.CtYBR Inv. 708(A) and (B): A Letter from the Prefect and a Court Summons.” *Zeitschrift für Papyrologie und Epigraphik* 219, 2021, pp. 161–166 [\[paper\]](#)

PROFESSIONAL EXPERIENCE

Salesforce — *Lead Data Scientist*

FEBRUARY 2025 – APRIL 2025

Led design and development of **text-to-SQL and RAG apps**; most impactful app projected to save **\$24M/year** in productivity

— *Data Scientist*

AUGUST 2022 – FEBRUARY 2025

Developed an end-to-end experimentation pipeline yielding a **15X increase** in Monthly Active Learners; co-built a **content recommendation engine** serving millions of users

— *Analyst, Strategy & Growth*

FEBRUARY 2021 – AUGUST 2022

Built out analytics and reporting program for Trailhead, a product with **>500K monthly active users**

SELECTED SKILLS AND AWARDS

Technologies: Python, PyTorch, Transformers, Scikit-Learn, SQL, AWS, Docker

Languages: Chinese (HSK 6); Spanish, French (working proficiency); German (limited); Latin, Ancient Greek (written)

Award: International Linguistics Olympiad, **first place worldwide** individually and team (2015)